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CSC 362 Data Visualization
May 13, 2026

Final Project: Movie Analysis

Dataset

We used data from the Github repository “Movie Revenue Analysis” by Natalya Doris (@ntdoris) in the file ‘final_dataset.csv,’ found at this [link](#). The dataset combines information from multiple sources (Box Office Mojo, IMDB, Rotten Tomatoes, TheMovieDB, and The Numbers) to create a comprehensive picture of how movie budget and revenue relate to other movie characteristics, such as genre.

The data is provided in a table, where each row represents a movie item. The attributes include movie/title, year, production budget, domestic gross (revenue), foreign gross (revenue), worldwide gross (revenue), month, profit, profit margin, roi (return on investment), pct foreign, match key, popularity, release date, original language, vote average, vote count, genre list, and each individual genre (19 total).

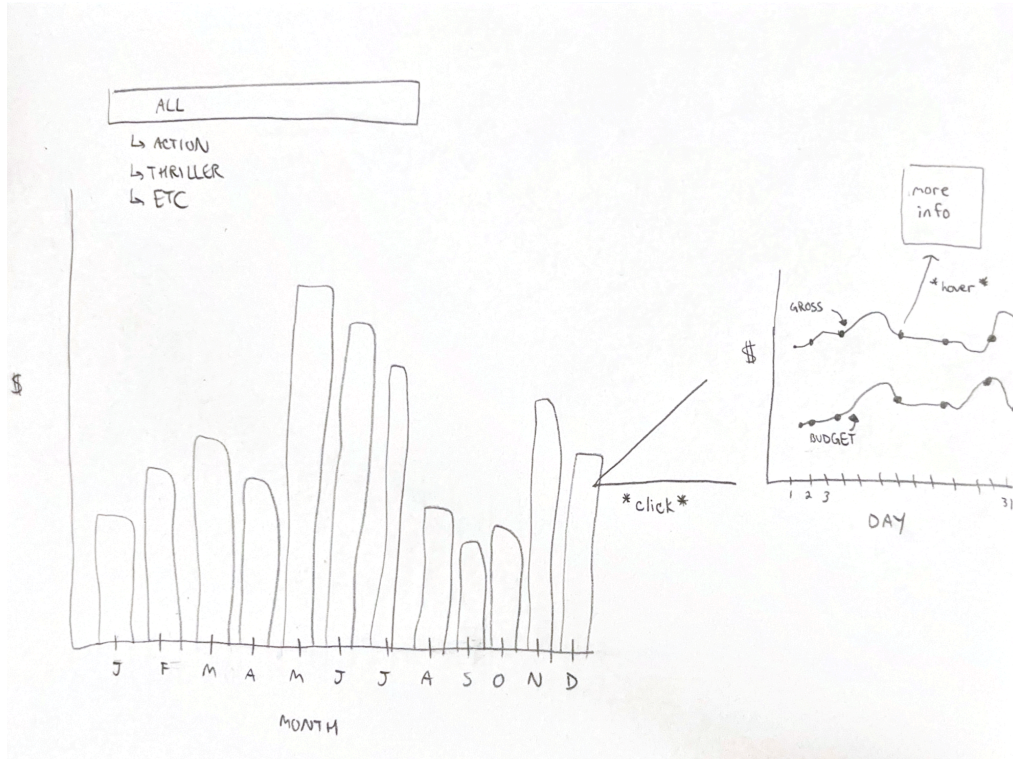
We found that the worldwide gross revenue matches the sum of domestic and foreign revenue, and the profit margin matches profit divided by worldwide gross revenue. For dates, the year and month are included as separate attributes, extracted from the release date column (which shows the full month, day, and year). For genres, the dataset includes one attribute with a list of all genres each movie falls under (for example “action, comedy, horror”), as well as a binary 0 or 1 for each separate genre attribute.

The dataset was complete with all information in all columns, so we did not have to handle null values. The student that compiled the dataset stated that she dropped rows that were missing genre information, which were a small percentage of the total dataset. She also filled in missing values in runtime data with the mean, which is not relevant to our project.

To clean our data, we first limited the columns to only include information relevant to our visualization. We kept the following attributes: movie/title, year, production budget, worldwide gross, month, profit, release date, genre list, and genres. Next, we wanted to focus on movies released in the 2010s, so we filtered the dataset using the year column to only include items labeled 2010-2019. After cleaning, the dataset we moved forward with included 1699 items.

Initial paper prototypes

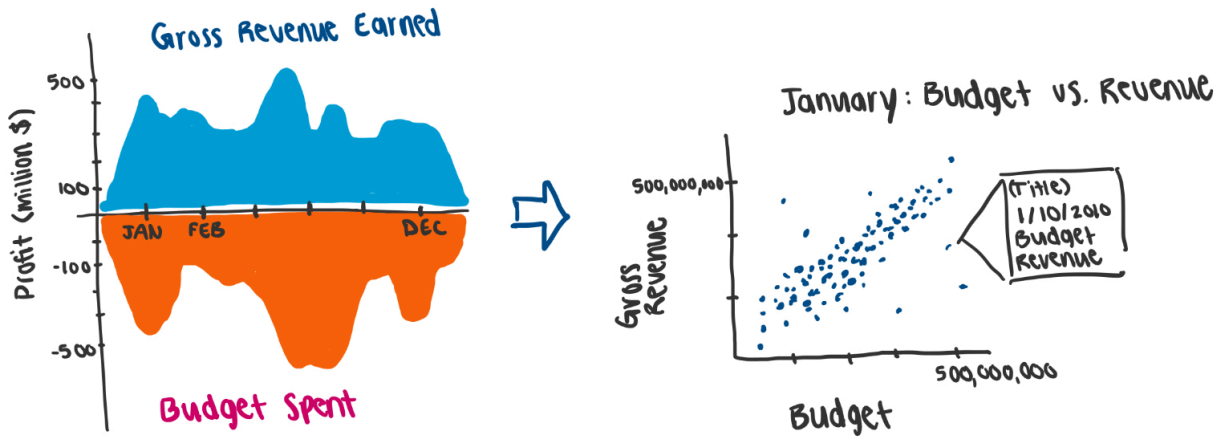
Prototype 1: Bar Chart & Line Graph



Prototype 1 uses a multifunction overview / detail coordination. The bar graph on the left shows the profit gained by movies released each month throughout the year (using all data). When a user clicks on the bar for a specific month, it shows the scatterplot on the right, allowing users to see the gross revenue and budget each day during a specific month (a subset of the data).

Users can click on specific points on the scatterplot to see more information in a tooltip, and will have the option to filter by genre (which applies to both visualizations). They will also be able to zoom in on each visualization (with constraints) to see the data for a specific month or day more clearly. This visualization does not use color as the primary channel to communicate any information, so colorblind users can easily understand it. We can utilize color as a second encoding to help differentiate between the revenue and budget lines in the graph, and use another color to fill in the bars. We also plan to use captions and alt text to explain each graph separately and what information it conveys.

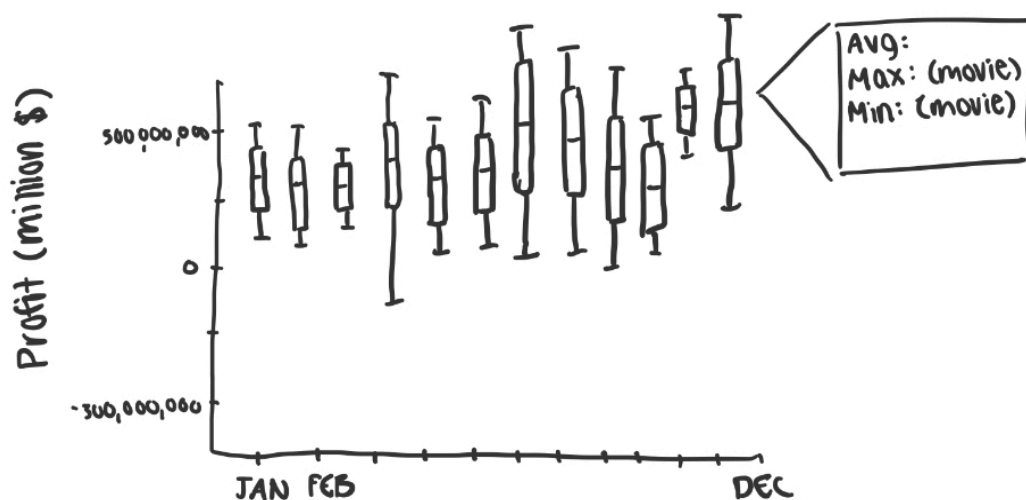
Prototype 2: Density Plot & Scatterplot



Prototype 2 uses different encodings with a similar coordination to Prototype 1. The density plot on the left shows both gross revenue and budget throughout the year. When a user clicks on a specific month, it shows the scatterplot on the right, which compares budget and revenue for each movie released during that month.

Users can also click on specific points on the scatterplot to see more information in a tooltip, have the option to filter by genre (which will apply to both visualizations), and zoom in on each visualization. Similar to Prototype 1, this visualization only uses color as a second encoding channel, so colorblind users do not miss any information. The revenue represents money gained, so I used a blue/green color and placed it above the x axis (positive). The budget represents money lost/spent, so I used a red/orange color and placed it below the x axis (negative). We also plan to use captions and alt text to explain each graph separately and what information it conveys.

Prototype 3: Box Plot



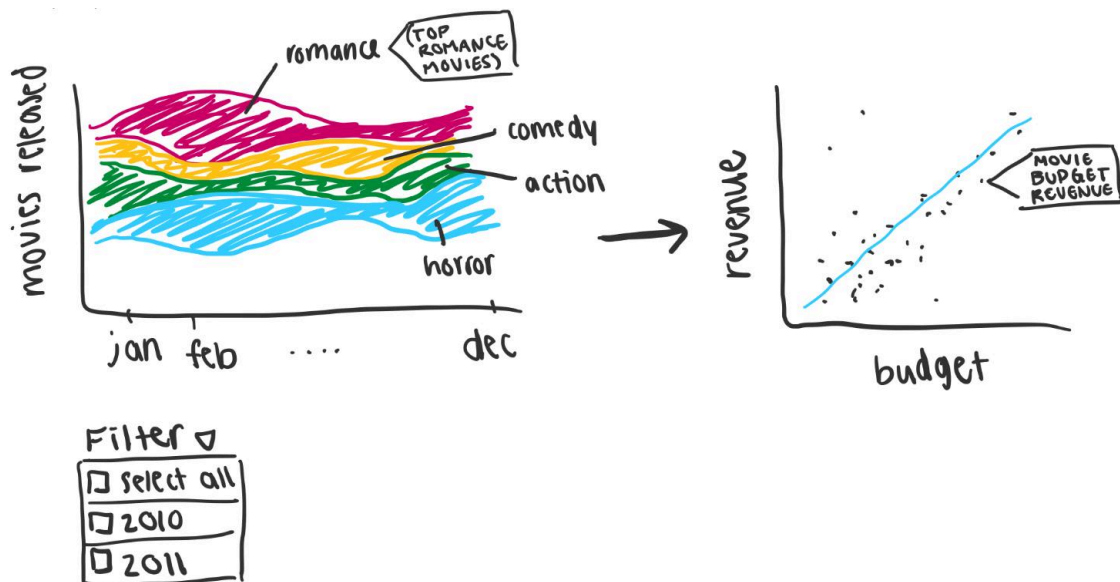
In Prototype 3, a box plot shows the profit minimum, maximum, and median for movies released each month throughout the year. When a user clicks on the box for a specific month, a tooltip gives more information about the summary statistics values and specific movies represented.

We plan to add the option to filter by genre, and could use this overview visualization with one of the detail view scatterplots from Prototype 1 or 2. Again, color is not necessary to understand this visualization, but we can use it as a second encoding channel for month or another attribute. We also plan to use captions and alt text to explain the graph and what information it conveys, and allow users to zoom in and out.

Revised Paper Prototype

After reviewing our prototypes, we thought more about the limitations of the dataset; we are not able to see during which specific months/days the revenue for each movie was generated, so our plan to show budget/revenue/profit by month/day did not make sense. Instead, we shifted our focus to see changes in how different genres are released throughout the year. We also used the scatterplot comparing revenue and budget from Prototype 2 to incorporate the financial attributes of the movie dataset (without relying on the release date).

Prototype 4: Streamgraph & Scatterplot



Prototype 4 includes a streamgraph showing how genres change throughout the year, paired with a scatterplot showing the relationship between movie budget and revenue within each genre. In the streamgraph, each color is a genre, and the thickness at a given month represents the number of movies released. To handle movies with multiple genres to ensure they don't count twice, we will divide each movie between its listed genres (for example, with 4 genres,

each would receive 0.25 points). When the user clicks on a stream, the scatterplot dataset will adjust accordingly.

We plan to add a filter by year so users can explore a specific year or use the full dataset, and users can also hover over/click on each stream and each point in the scatterplot to see a tooltip with more information. Users will be able to zoom in on the scatterplot (with constraints), however we don't think zooming adds any benefit to the streamgraph. Color is necessary to interpret the stream graph, so we will have to consider repeating colors and ensure there is high enough contrast between adjacent streams. We also plan to incorporate keyboard navigation to allow users to move through and select streams, and will add captions and alt text to explain the graph and what information it conveys.

Task & Accessibility Analysis

Task 1: Compare – In December, was a greater share of horror or comedy movies released?

- A general user can identify December on the x axis, find the streams for horror and comedy, and visually compare which is thicker. The size channel conveys this attribute (share of movies) for each area mark (stream).
- A colorblind user can find the text labels “horror” and “comedy” on the streamgraph. If this is difficult, they can also hover over the text in the legend to highlight the corresponding stream. If a user is unable to navigate with a mouse, they can use the built-in keyboard navigation feature by clicking Tab to enter the stream graph, then using the right/left arrow keys to move between genre streams. If users are unable to see a difference visually, they can hover over each stream to see the numbers of shares in the tooltip, and compare which value is greater.
- It is important to note that the tooltip text summarizes the genre across all months, so fully visually-impaired users might miss smaller patterns across different months. Further, although most visual information is also communicated through other tools and features, it is more time consuming to access through non-visual methods.

Task 2: Range – What is the range of thriller movie budgets?

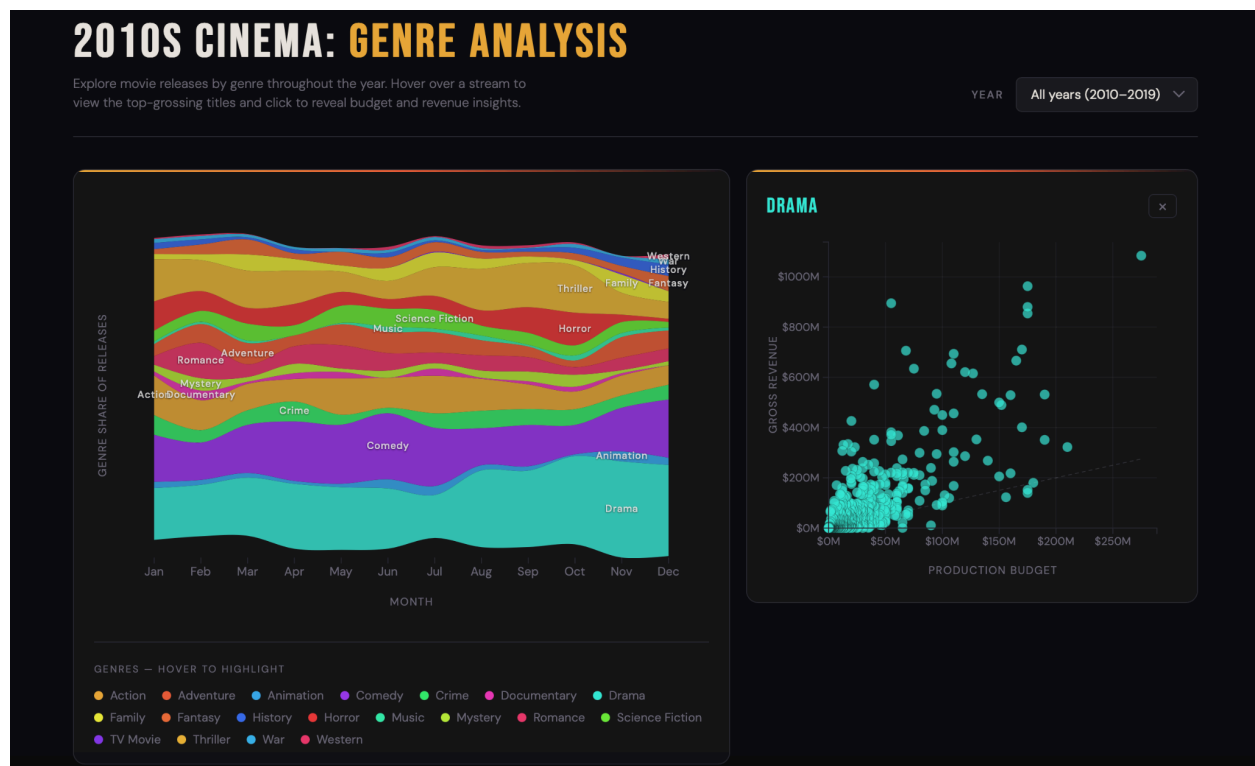
- Once the thriller scatterplot is open, a general user can identify that the budget is on the x axis. The point marks use the spatial position channel, with horizontal position representing budget. Users can find the furthest point on the left and right. For visually impaired users (or if visually identifying spatial position is difficult), they can hover over a point to see a tooltip with text that includes the specific budget value. This also provides more precise information, compared to a range estimated visually.
- The user can access all relevant information within the tooltip text, however the process is more time consuming without visual information to guide the user's actions.

Task 3: Order – Order all Western movies from least to greatest in terms of the gross revenue they attained.

- After navigating to the Western scatterplot, users can identify that revenue is on the y axis. The vertical spatial position channel encodes this attribute for each point mark (representing one movie), so the user can roughly order the points by their vertical position. Since the points are not aligned (given the x position encodes another attribute), this might be difficult if two points are relatively close, especially for visually-impaired users. In this case, users can open the tooltip to confirm the specific values, which provides more precise information.
- Similar to the previous tasks, it is possible for users that are unable to use the visual spatial position to rely solely on the text and tooltip information, however the process is slower. This task would also be more difficult for other genres that contain more data points (Western only includes 16 movies).

Class Demos & Feedback

First iteration:



After receiving feedback during class demos, we made the following revisions to our initial visualization:

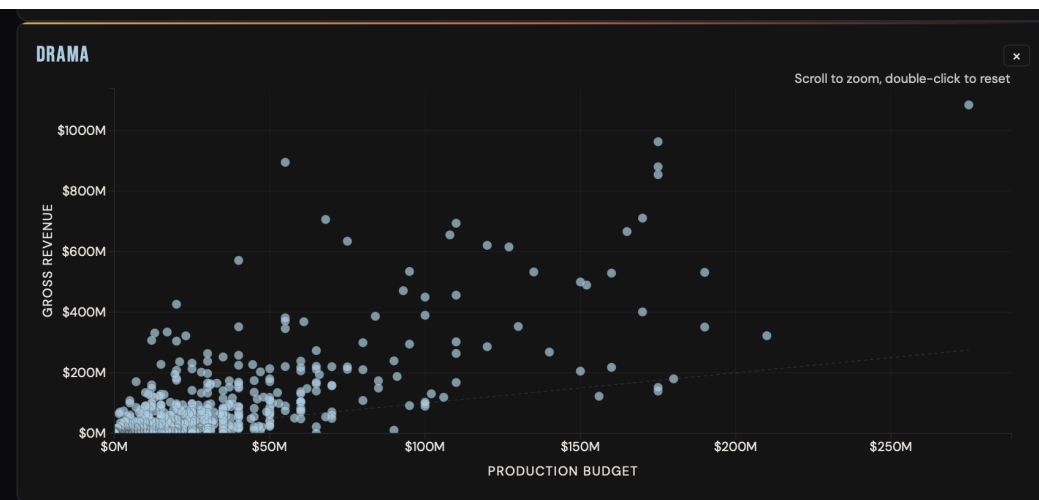
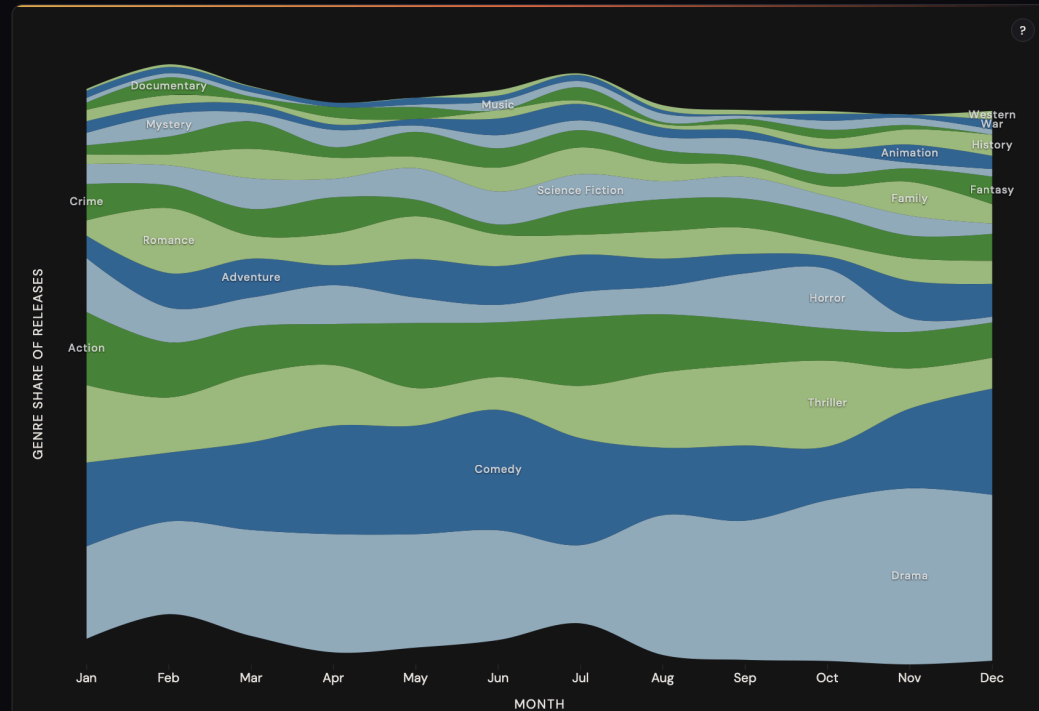
- Standardized the vertical order of streams to remain consistent through filtering
- Utilized color brewer to check that the streamgraph color palette is colorblind safe – switched to a maximum of 4 colors that is ideal for qualitative data (light green, dark green, light blue, and dark blue) and repeated the pattern
- Made smaller streams more visible by increasing the height of the visualization and adding a hover opacity to the genre legend
- Increased the font size and used a lighter color for higher contrast against the black background to increase text accessibility
- Added a zoom feature to the scatterplot because users found that it was hard to see individual points in the clusters around the bottom left corner
- Found that 2019 only included 3 movies and lacked revenue data, so removed this column and chose to limit analysis to 2010-2018
- Found few movies in the TV Movie genre, and users were unsure what this category meant, so chose to remove it
- Added the ability to click on the genre word in the legend (in addition to the genre stream itself) to see the scatterplot
- Added keyboard navigation to move through the streams
- Users found it hard to quantify the streamgraph values visually, so added the total share of movies in that genre to the tooltip
- Added a help button to view a popup text box explaining how to use and interpret the visualizations, filters, and other features
- Added a link to the original dataset
- Added a constraint to the position of scatterplot tooltip so it always stays on screen (previously got cut off for points on the far right of the scatterplot)
- Adjusted the labels on the streamgraph and axes titles so no text overlaps
- Switched from a horizontal to vertical layout to allow users to see each visualization more clearly (and users can still zoom out to see both simultaneously)

Revised visualization:

2010S CINEMA GENRE ANALYSIS

Explore movie releases by genre throughout the year. Hover over a stream to view the top-grossing titles and click to reveal budget and revenue insights.

YEAR All years (2010–2018) ▾



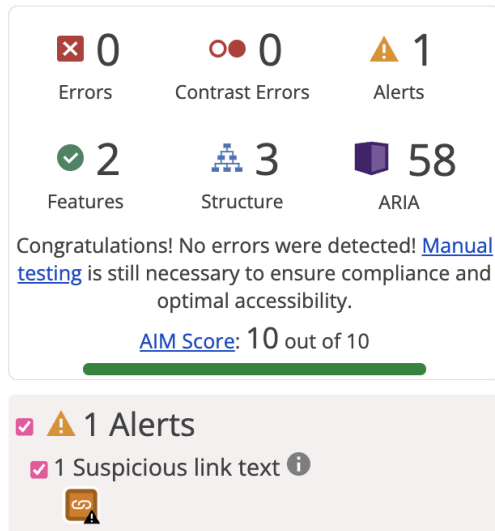
View the original dataset in the Github repository "[Movie Revenue Analysis](#)" by Natalya Doris (@ntdoris)

Accessibility Check

We completed the WAVE test to see if a user can accomplish our tasks with a screenreader. We achieved the following results:

- 0 errors
- 1 contrast error – very low contrast on the empty scatterplot icon and background (fixed)
- 0 alerts
- WAVE Accessibility Impact Score: 9.7/10

WAVE results after fixing the contrast error:



Pilot Testing

Next, we conducted pilot testing with a student participant to identify areas to improve our visualization for the final iteration. We used the following procedure to conduct the experiment:

- A speaker, notetaker, and participant should be present
- Provide the participant with access to the visualization and a list of the tasks to refer back to
- The speaker follows a script to guide the participant through the testing process
- The notetaker observes the participant & records notes on how they complete each task
- The speaker and notetaker review the participant's feedback and identify changes to make in the visualization in order to address any problems identified during the pilot testing process

Speaker script:

Speaker: "We are evaluating our visualization and are asking you, the participant, to complete some tasks using the visualization and then provide feedback about your experience. As a reminder, we are evaluating the visualization, not you as a participant, so you don't need to worry about being "right" as you complete these tasks. There are three tasks, followed by a brief feedback session. The whole pilot session should take under 5 minutes. Do you consent to participate?"

[Wait for yes]

Speaker: “Thank you for agreeing to participate. We will start with the three tasks. Please ‘think aloud’ as you complete the task, meaning voice what you are thinking as you work through the task. Please talk through your full thought process, including parts you think are easy and any challenges you encounter. Feel free to ask clarifying questions. Your first task is: In December, was a greater share of horror or comedy movies released?”

- * The speaker may answer clarifying questions, such as “what do you mean by x?” or “are you looking for an exact number or an approximation?”, but should avoid providing any assistance to the user in completing the task

[Pause to allow the participant to complete the task]

Speaker: Please continue ‘thinking aloud.’ The second task is: What is the range of thriller movie budgets?

[Pause to allow the participant to complete the task]

Speaker: Your third task is: Order all Western movies from least to greatest in terms of the gross revenue they attained.

[Pause to allow the participant to complete the task]

Speaker: “That is the end of the third task. Now, we welcome any feedback you may have about the visualization or about your process for completing the tasks. Which parts were the most straightforward for you, and which were the most confusing?”

[Allow participant to answer]

Speaker: “Are there any resources or features you wished the visualization had that would have helped you complete the tasks?”

[Allow participant to answer]

Speaker: “Do you have any other feedback or ideas you would like to share?”

[Allow participant to answer]

Notes on how the participant completed each task:

- Task 1 -
 - Initially, participant was not sure what a “share” means (more confusion over the wording of the task than the visualization itself)
 - She was a little confused when first viewing the visualization, and had to take a few seconds to get oriented and understand what each visualization was showing
 - She was able to locate the correct streams on the graph (comedy and horror) and find December at the righthand side of the x axis
 - She correctly assumed that the thickness of the stream represented the share of movies, and correctly stated that the comedy stream was thicker than horror in December, meaning the share of movies was greater
- Task 2 -
 - At first, the participant didn’t realize she could scroll down further to see another visualization below the streamgraph
 - After finding the scatterplot, she was easily able to understand that clicking on a genre stream changed the values in the scatterplot, and identified and clicked the thriller section
 - In the scatterplot, she saw that the x axis represented budget, with increasing values from left to right

- She correctly stated the maximum budget by hovering over the furthest right point and reading the value in the tooltip
- She first estimated the minimum value to be \$0, which is how it appears visually from the fully zoomed out graph. She then revised her answer to the exact answer, \$25,000, after hovering over the point to see the tooltip. She did not realize she could zoom in to see the points in the bottom left more clearly, which was not essential but could have helped with this task.
- Task 3 -
 - At this point, the participant had a better understanding of how the two visualizations worked, both separately and together, and was quickly able to find and click the Western stream to change the scatterplot.
 - She correctly identified that the spatial position along the y axis represented the revenue, increasing from bottom to top.
 - She ordered the movies correctly by estimating their visual position, then checked her answers by hovering over the points to confirm the exact values in the tooltip popup.
- Overall, the participant was a little confused initially, but was able to fully complete every task. She asked questions and explained points of confusion throughout, but was consistently able to problem-solve any issues independently without additional help from the speaker/notetaker.

Participant feedback:

- The visualization was a little overwhelming initially. The participant stated that the hardest part was understanding what was happening when first viewing the website, but she understood it better and felt more comfortable as she explored and interacted more with the visualization.
- The participant suggested making everything in the streamgraph a different color. She pointed out that shared colors imply a relationship (for example, that all light blue sections of the streamgraph are related). This makes sense, but is difficult to balance with making the visualization colorblind friendly.

Changes to make:

- Make the help popup window present when a user first enters the website (then they can close it and reopen it any time).
 - The popup is currently only accessible through a button in the top corner, which the participant didn't see. The information in the popup window would have answered all of the questions she asked as she completed the tasks. This would help the participant be aware of several components she initially missed, such as the second graph below the first and the ability to zoom on the scatterplot.
 - This change would also address one of her main points of feedback – that the hardest part was first understanding the visualization. Providing information first might make it more approachable to users that have no background knowledge and might benefit from guidance on how to interpret the graphs and utilize the interactive features.

- Make the instructional text (specifically on how to interact with the streamgraph and how to zoom on the scatterplot) slightly bigger so it stands out more.

Final Revisions

We read through the Chartability Tests, WAVEd our website, and completed all accessibility checks on our visualization.

- Standards we confirmed our visualization met – high contrast, signifiers on how to use visualizations (highlighting, mouse symbol changes on interactive components), tasks can be accomplished on 400% zoom (more difficult and requires a lot of scrolling, some text overlaps, but possible to complete tasks successfully), keyboard navigation (only on the streamgraph – chose not to add this feature to the scatterplot given some have over 100 points), “help icon” that provides the user with more info on how to view the visualization
- New changes we made included:
 - Adding a link to the data in an HTML table format
 - Ensuring all font sizes are 12px minimum
 - Editing wording of title and other texts
 - Adding visible caption with main takeaways
 - Adding text description visible to screen reader with more details for each vis
 - Adding a descriptive title for each vis
 - Adding the ability for users to tab through sections of the website
 - Editing the tooltip to no go off the edges of the screen

Access the final visualization [here](#).

We heavily relied on AI for the tasks we didn’t have as much exposure to in class, particularly the stream graph. We began by prompting Claude to give us boilerplate code for a first draft of the stream graph. The Lab 13 vibecoding exercise proved particularly helpful here, as it helped us maintain enough specificity in our prompts to get useful output. We used a similar approach as we have in the labs by using a provided code outline, then working to understand each part of it as we edited different components ourselves.

For the other steps—editing the stream graph, linking the scatterplot, adding the dropdown filter, implementing hover and click interactivity, creating the help icon button, etc—we referenced previous labs (mainly labs 5,7, and 8), Stack Overflow, W3Schools, and Google’s AI Overview as needed. The scatterplot zoom feature was where we felt the most lost, and we leaned heavily on Claude to work through creating that feature. Claude also provided the large majority of our CSS styling, which gave the visualization its cinema-themed aesthetic. Throughout the project, we also used ChatGPT to assist with specific error messages and debugging when necessary. We revised the visualizations, text, and style on our own, as well as evaluating the visualization for accessibility and adding additional features throughout.

User testing

We followed the same procedure and script outlined above in Pilot Testing with three new participants.

Participant 1 (Environmental Science & Political Science double major)

Notes on how the participant completed each task:

- Task 1 -
 - The participant quickly read over the instructions in the popup window
 - She hovered over multiple sections of the streamgraph to see the different values in the tooltip, then located the horror and comedy streams
 - She saw the months along the x-axis and located December, then was easily able to determine that the comedy stream was thicker, meaning a greater share
- Task 2 -
 - The participant initially looked for budget information in the streamgraph and was confused why it wasn't included before realizing she could scroll down to see the scatterplot
 - Once she found the scatterplot, she used the legend text to select the thriller genre
 - She was familiar with a scatterplot graph format, so she was easily able to find budget along the x axis and identify the minimum and maximum points
 - The participant estimated the values at \$275 million (correct) and \$0 (relatively close) based on the points' horizontal spatial position without using the tooltip
- Task 3 -
 - The participant was familiar with navigating the scatterplot, so she was able to see revenue on the y-axis and ordered the movies from the top to bottom vertical spatial position
 - She realized that she could see the movie title and exact values in the tooltip by hovering over the point, which she had read in the initial instructions but forgot she could do

Participant feedback:

- The participant stated that she found everything very straightforward
- She found it helpful to read over the instructions popup at the beginning
- She thought the colors looked good aesthetically and were easy to differentiate
- She also thought it had a good balance of text and visuals, and all text was very clear and easy to read. Specifically, she found that the legends and axes made the graphs easy to understand and interpret
- The participant's main point of confusion was that she forgot there were two visualizations, and initially assumed she could find information to complete all tasks within the streamgraph (she was confused starting Task 2 when the streamgraph did not include financial information)

Participant 2 (Classics & History double major)

Notes on how the participant completed each task:

- Task 1 -
 - The participant skimmed through the instructions, then started hovering over different streams in the scatterplot
 - She then quickly found the month December on the x axis, and located the streams for horror and comedy
 - She correctly answered that comedy has a greater share in December
- Task 2 -
 - The participant first looked for budget information in the tooltip of the streamgraph before scrolling down to find the scatterplot
 - She found that points' position along the x axis represented the budget
 - She identified the maximum point on the right and opened the tooltip to find the exact value, \$275 million. She then estimated the minimum to be under \$1 million, before hovering over the minimum point to find that the exact value was \$25,000. She did not use the zoom function, which possibly could have made this task easier.
- Task 3 -
 - At this point the participant had an understanding of how to use the scatterplot and tooltip, so she was able to complete this task quickly. She found the budget label on the y axis, and started at the maximum point. She moved her way down through the points, using the tooltip to find each movie title and check the exact budget values.

Participant feedback:

- The participant thought that the visualization was relatively straightforward, and she was able to complete each task relatively quickly.
- She thought the instructions were helpful, but provided a large amount of information/text that she only skimmed; she pointed out that if she read it more in-depth, it could have helped her complete the tasks quicker.

Participant 3 (Environmental Science & Biology double major)

Notes on how the participant completed each task:

- Task 1 -
 - The participant fully read through the instructions, then scrolled through the full visualization and hovered to find both tooltips before starting on the task
 - She located the streams for horror and comedy, then found where they lined up on the far right side with December on the x axis
 - She correctly answered that the comedy stream is thicker, so it has a greater share
- Task 2 -
 - The participant found the budget label on the x axis, and correctly assumed that points' horizontal position showed their budget, increasing from left to right
 - She identified the points positioned furthest to the left and right, and was easily able to open the tooltip to find the exact values, \$275 million and \$25,000.

- She struggled a little to find the exact minimum point as there was a cluster on the left side of the graph, but it only took a few seconds. She did not use the zoom function, which possibly could have made this part easier.
- Task 3 -
 - Similar to the other participants, this task was the quickest, likely because she had an understanding of how to navigate the visualization and what information was included. She followed the same process as in task 2, instead finding the budget label on the y axis and opening the tooltip for each point. She started at the maximum point at the top of the graph, and moved down to the minimum at the bottom.

Participant feedback:

- The participant found the information at the beginning very helpful, but didn't realize she could refer back to it after with the help button.
- Overall she found all the tasks easy to complete, and thought the visualization looked nice, was informative, and was relatively straightforward to navigate. She also liked the interactive components, which made it more engaging. She did not have any specific constructive feedback.

Summary

- Overall, each participant faced minor points of confusion, but was able to complete every task independently and relatively quickly.
- All participants stated that they found the visualization easy to understand, especially as they interacted with it more. Two skimmed the instructions at the beginning, which they found helpful but thought was a lot of text, so they did not fully read it before closing the window to see the visualization itself. Only one participant read through the full instructions, but was able to complete the tasks with less difficulty.
- Two out of three participants initially didn't realize there were two connected visualizations, which was the main challenge they faced (at the beginning of Task 2). If we had more time to work on this project, we would add a feature to indicate that users can scroll past the streamgraph to see the scatterplot, and make the help icon more obvious. Beyond this, there were no major challenges or points participants got stuck on.